

Market Liquidity and Funding Liquidity*

Markus K. Brunnermeier[†]
Princeton University

Lasse Heje Pedersen[‡]
New York University

March 2005

Abstract

We provide a model that links a security's market liquidity — i.e., the ease of trading it — and traders' funding liquidity — i.e., their availability of funds. Traders provide market liquidity and their ability to do so depends on their funding, that is, their capital and the margins charged by their financiers. In times of crisis, reductions in market liquidity and funding liquidity are mutually reinforcing, leading to a liquidity spiral. The model provides a natural explanation for the empirically documented features that market liquidity: (i) periodically dries up, (ii) has commonality across securities, (iii) is related to volatility, (iv) experiences “flight to liquidity” events, and (v) comoves with the market. Finally, the model shows how the Fed can improve current market liquidity by committing to improve funding in a potential future crisis.

Keywords: Risk Management, Liquidity, Liquidation, Systemic Risk, Margins, Haircuts

*Preliminary. Please do not distribute. We are grateful for helpful comments from Yakov Amihud, David Blair, John Kambhu, and Dimitri Vayanos. We would also like to thank seminar participants at the New York Federal Reserve Bank and the New York Stock Exchange.

[†]Princeton University, Department of Economics, Bendheim Center for Finance, Princeton University, 26 Prospect Avenue, Princeton, NJ 08540-5296, e-mail: markus@princeton.edu, <http://www.princeton.edu/~markus>

[‡]New York University and National Bureau of Economic Research (NBER), 44 West Fourth Street, NY 10012-1126, e-mail: lpederse@stern.nyu.edu, <http://www.stern.nyu.edu/~lpederse/>

1 Introduction

Trading requires capital. When a trader — e.g. a dealer, hedge fund, or investment bank — buys a security, he can use the security as collateral and borrow against it, but he cannot borrow the entire price. The difference between the security’s price and collateral value, denoted the margin or the haircut, must be financed with the trader’s own capital. Similarly, shortselling requires capital in the form of a haircut; it does not free up capital. Hence, at any time the total haircuts on all positions cannot exceed the trader’s capital.

Our model shows that the funding of trades affects, and is affected by, market liquidity in a profound way. When funding liquidity is tight, traders become reluctant to take on positions, especially “capital-intensive” positions in high-haircut securities. This lowers market liquidity, especially for high-haircut securities. Further, low future market liquidity increases the risk of financing a trade, thus increasing the haircut. Based on the links between funding and market liquidity, we provide a unified explanation for the main empirical features of market liquidity. In particular, the model implies that market liquidity (i) periodically dries up, (ii) has commonality across securities, (iii) is related to volatility, (iv) experiences “flight to liquidity” events, and (v) comoves with the market.

Our model is similar in spirit to Grossman and Miller (1988) with the new feature that dealers face the real-world funding constraint discussed above. In our model, a group of “initial customers” face a supply shock at time one, which affects their willingness to hold shares. A group of “complementary customers” face the opposing shock, but these agents arrive in the market only at a later time T . A group of dealers bridge the gap between the initial and complementary customers by smoothing the price and thus providing market liquidity.

We derive the competitive equilibrium of the model and define precisely both liquidity concepts within this context. Specifically, market liquidity is defined, for each trade size, as the difference between the transaction price the fundamental value; funding liquidity is defined as a dealer’s marginal value of wealth, which is shown to equal one plus the current and expected future Lagrange multipliers on the capital constraint.

With these definitions, we can explore the model’s liquidity implications. Naturally, as long as dealer capital is abundant, market liquidity is at its highest level and insensitive to marginal changes in capital and haircuts. However, when dealers hit capital constraints then they are forced to reduce their positions and market liquidity is reduced. Further, in this case market liquidity is sensitive to changes in dealer wealth and to financing conditions; thus market liquidity is volatile in times of crisis.

Since market making firms often are long the market, capital constraints are more likely to be hit during market downturns. When capital is tight and the market becomes illiquid, market liquidity further becomes sensitive to changes in funding conditions including traders’ capital and haircuts. Hence, the model implies that market liquidity

can suddenly dry up, and can help explain why this is likely to happen in a down market.

Further, the model implies that market liquidity dry-ups can be erratic because of inherent amplification mechanisms. If dealers have large positions when they hit a capital constraint then a price drop causes dealer losses, leading to dealer sales and further price drops, and so on. This “backward bending” dealer demand implies, paradoxically, that a shock to the customers’ demand for immediacy leads to a reduction in the provision of immediacy in such stress times.

Our model also provides a natural explanation for the commonality of liquidity across assets since shocks to the funding constraint of the dealer-sector affects all securities. This may help explain that market liquidity is correlated across stocks as documented by Chordia, Roll, and Subrahmanyam (2000), Hasbrouck and Seppi (2001) and Huberman and Halka (2001), and across stocks and bonds as documented by Chordia, Sarkar, and Subrahmanyam (2005).

Next, our model predicts that market liquidity declines with fundamental volatility, which is consistent with the empirical findings of Benston and Hagerman (1974) and Amihud and Mendelson (1989). To see the intuition for this result, note first that fundamental volatility trivially leads to price volatility, which leads to higher haircuts. Consequently, it is more capital intensive for dealers to trade in volatile securities and, therefore, dealers provide less market liquidity in such securities. The reduced market liquidity further increases the risk of financing such trades, thus further increasing haircuts, and so on. This reasoning applies both when comparing the market liquidity across securities in the cross section, and when explaining changes in market liquidity in the time series.

The model implies that the liquidity differential between high-volatility and low-volatility securities increases as dealer capital deteriorates — a phenomenon often referred to as “flight to quality” or “flight to liquidity”. According to our model, this happens because a reduction in dealer capital induces traders to provide liquidity mostly in securities that do not use much capital, that is, low-volatility securities, which have lower haircuts. Acharya and Pedersen (2005) document empirical evidence consistent with flight to liquidity.

Our model also has implications for the optimal management of funding liquidity risk — the risk that the funding constraint becomes binding. Dealers try to reduce the risk of forced liquidation by cutting back their positions earlier and by maintaining a “safety buffer.” This buffer trades off the profit from investing early against the cost of funding problems in the future. In our multiple-asset setting (risk-neutral) traders also take cross-correlations among assets’ fundamentals and future market liquidities into account, since well-diversified positions reduce funding liquidity risk.

Our model thus provides insights on the interaction of funding liquidity and market liquidity, and can help explain the major empirical features associated with market liquidity. Furthermore, the model suggest a novel line of empirical work, namely to

link empirically measures of dealer’s funding to measures of market liquidity.

The remainder of the paper proceeds as follows. Section 2 describes the institutional features associated with the financing of trading activity for marketmakers, banks, and hedge funds, and discusses funding liquidity risk. Section 3 present our formal model. Section 4 characterizes the main features of equilibrium, and Section ?? derives the model’s liquidity implications. Section 5 shows how our framework relates to the liquidity concepts discussed within the literatures on market microstructure, corporate finance, banking, limits of arbitrage, and macroeconomics. The appendix contains proofs.

2 The Funding Liquidity of Traders

There are several types of providers of market liquidity, that is, traders that act as intermediaries by buying or selling. The main types of such traders are market makers, proprietary traders, and hedge funds. These traders are subject to funding constraints on the their trading activity, and we refer to the risk of a binding funding constraint as funding liquidity risk. To set the stage for our model, we review the main real-world funding constraints for securities firms.

2.1 Haircuts and Capital Constraints: The Case of Hedge Funds

We consider first consider the funding issues faced by hedge funds since they have relatively simple balance sheets and face little regulation. Below, we discuss the funding issues (including regulation) for banks, and market makers.

A hedge fund must finance its activities using its capital. A hedge fund’s capital W_t at time t consists of its equity capital supplied by the partners and of possible long-term debt financing that can be relied upon during a potential funding crisis. Since a hedge fund is a partnership, the equity is not locked into the firm indefinitely as in a corporation. The investors (that is, the partners) can withdraw their capital at certain times, but the withdrawal is subject to so-called lock-up periods (typically several months or even years) to ensure the funding. A hedge fund usually cannot issue long-term unsecured bonds, but some (large) hedge funds manage to obtain debt financing in the form of medium term bank loans or in the form of a guaranteed line of credit.¹ Hedge funds lever their capital using collateralized borrowing financed by the hedge fund’s prime broker(s). The prime brokerage business is opaque since the terms

¹A line of credit may have a “material adverse change” clause or other covenants subject to discretionary interpretation of the lender. Such covenants imply that the line of credit may not be a reliable source of funding during a crisis.

of the financing are subject to negotiation and are secret to outsiders. We describe stylized financing terms and, later, we discuss caveats.

If a hedge fund buys at time t a long position of $x_t^j > 0$ shares of a security j with price p_t^j , then this requires the hedge fund to come up with $x_t^j p_t^j$ dollars. The security can, however, be used as collateral for a new loan of say l_t^j dollars. The difference between the price of the security and the collateral value is denoted the haircut (or, margin requirement) $h_t^{j+} = p_t^j - l_t^j$. Hence, this position uses $x_t^j h_t^{j+}$ dollars of the fund's capital. We note that the collateralized funding implies that the capital use depends on haircuts, not notional amounts.

The haircuts on fixed income securities and over-the-counter (OTC) derivatives are set through a negotiation between the hedge fund and the broker that finances the trade, often the hedge funds' prime broker. The haircut is typically set such as to make the market almost risk free for the broker. Hence, the collateral value of a long position l_t^j for borrowing between time t and $t + 1$ is the smallest possible value that the security might have to be sold for at time $t + 1$ (with a certain confidence) in case the borrower defaulted on the loan.²

In the U.S., haircuts on equities are subject to Regulation T, which stipulates that all non-broker/dealers must have an initial haircut (initial margin) of 50% of the market value of the underlying stock, both for long and short positions. Hedge funds can circumvent Regulation T for instance by organizing the transaction as a total return swap, which is a derivative that is functionally equivalent to buying the stock.

The haircut on exchange traded futures (or options) is set by the exchange. The principle for setting the haircut for futures or options is the same as that described above: The haircut is set such as to make the exchange almost immune to default risk of the counterparty, and hence riskier contracts have larger haircuts.

At the end of the financing period, time $t + 1$, the position is "marked-to-market," which means that the hedge fund receives any gains (or pays any losses) that have occurred between t and $t + 1$, that is, the fund receives $x_t^j(p_{t+1}^j - p_t^j)$, and the fund pays interest on the loan, $r_t x_t^j l_t^j$, where r_t is the funding rate. If the trade is kept on, the broker keeps the haircut to protect against losses going forward from time $t + 1$. The haircut can be adjusted if the risk of the collateral has changed unless the counterparties have contractually fixed the haircut for a certain period.

If the hedge fund wants to sell short a security, then the fund asks one of its brokers to locate a security that can be borrowed, and then the trader sells the borrowed security.³ The broker requires a collateral that we denote by c_t^j . The collateral value of a short position c_t^j is the highest possible value that the security might have to be

²This level of the haircut is sometimes referred to as the broker's "potential future exposure" (PFE). Often brokers also take into account the time it takes between a fail by the hedge fund is noticed and the security is actually sold. Hence, the haircut of a one-day collateralized loan depends on the estimated risk of holding the asset over a time period that is often set to be five to ten days.

³See Duffie, Gârleanu, and Pedersen (2002).

bought back for at time $t + 1$ (with a certain confidence). As with a long position, this makes the transaction almost risk free for the broker. This collateral requirement implies that the short sale uses $x_t^j(c_t^j - p_t^j)$ of the hedge fund's capital. The haircut on a short position — i.e. the per share capital use — is denoted by $h_t^{j-} = c_t^j - p_t^j$. We note that a short sale does not raise capital for a hedge fund; it uses capital.

A hedge fund must be able to finance all its security positions at any point of time. This means that the total capital use must be smaller than the available net capital plus available, that is, at any time t ,

$$\sum_j (x_t^{j+} h_t^{j+} + x_t^{j-} h_t^{j-}) \leq W_t \quad (1)$$

where $x_t^{j+} \geq 0$ and $x_t^{j-} \geq 0$ are the positive and negative parts of $x^j = x_t^{j+} - x_t^{j-}$, respectively.

Our description of the determination of haircuts was described as if haircuts are set separately for each security position. It is, however, sometimes possible to jointly finance several trades that are part of the same strategy. This leads to a lower total haircut if the risks of the various positions are partially offsetting. For instance, much of the interest rate risk is eliminated in a “spread trade” with a long position in one bond and a short position in a similar bond. Hence, the haircut of a jointly financed spread trade is smaller than the sum of the haircuts of the long and short bonds. For a strategy that is financed jointly, we can reinterpret security j as such a strategy. Prime brokers compete by, among other things, offering low haircuts — a key consideration for hedge funds — which means that it is becoming increasingly easy to finance more and more strategies jointly. In the extreme, one can imagine a joint financing of a hedge fund's total position such that the funding constraint becomes

$$m_t(x_t^1, \dots, x_t^J) \leq W_t \quad (2)$$

where m is the margin requirement of the portfolio x , that is, the most the portfolio can lose over the next funding period with a certain confidence. This is a Value-at-Risk (VaR) constraint as discussed further in Section 2.4. Currently, it is often not practical to jointly finance a large portfolio. This is because a large hedge fund finances its trades using several brokers, a hedge fund and a broker each can consist of several legal entities (possibly located in different jurisdictions), certain trades need separate haircuts paid to exchanges (e.g. futures and options) or to other counterparties of the prime broker (e.g. securities lenders), prime brokers may not have sufficiently sophisticated models to evaluate the diversification benefits (e.g. because they don't have enough data on the historical performance of newer products such as CDOs), and because of other practical difficulties in providing joint financing. Further, if the margin requirement m relies on assumed stress scenarios in which the securities are perfectly correlated (e.g. due to predatory trading), then (2) coincides with (1).

2.2 Funding Requirements for Banks

A bank's capital W consists of equity capital plus its long-term borrowings (including credit lines secured from individual or syndicates of commercial banks), reduced by assets that cannot be readily employed (e.g. goodwill, intangible assets, property, equipment, and capital needed for daily operations), and further reduced by uncollateralized loans extended by the bank to others (e.g. Goldman Sachs 2003 Annual Report). Banks also raise money using short-term uncollateralized loans such as commercial paper and promissory notes, and, in the case of commercial banks, demand deposits. These sources of financing cannot, however, be relied on in times of funding crises since lenders may be unwilling to continue lending, and therefore these short term funding is not included in W .

The financing of a bank's trading activity is largely based on *collateralized* borrowing. Banks can borrow securities to short from mutual funds and pension funds, for instance, and can finance long positions using collateralized borrowing from corporations with excess cash, other banks, insurance companies, and the Federal Reserve Bank. These transactions typically require haircuts which must be financed by the bank's capital W as captured by the funding constraint (1).

The financing of a bank's proprietary trading is more complicated than that of a hedge fund, however. For instance, banks may negotiate zero haircuts with certain counterparties, banks can often sell short shares held in house, that is, held in a customers margin account (i.e. held in "street name") such that bank does not need to borrow the share externally. Further, a bank receives haircuts when financing hedge funds (i.e. the haircut is negative from the point of view of the bank). However, often the bank wants to pass on the trade to an exchange or another counterparty and hence has to pay a haircut to the exchange. In spite of these caveats, we believe that in times of stress, banks face haircuts and are ultimately subject to a funding constraint in the spirit of (1) (see, e.g., Goldman Sachs, 2003 Annual Report, page 62).

In addition, banks have to satisfy certain regulatory requirements. Commercial banks are subject to the Basel accord, supervised by the Federal Reserve system for US banks. In short, the Basel accord of 1988 requires that a bank's "eligible capital" exceeds 8% of the "risk-weighted asset holdings," which is the sum of each asset holding multiplied by its risk weight. The risk weight is 0% for cash and government securities, 50% for mortgage-backed loans, and 100% for all other assets. The requirement posed by the 1988 Basel accord corresponds to Equation (1) with haircuts of 0%, 4%, 8%, respectively. In 1996, the accord was amended, allowing banks to measure market risk using an internal model similar to (2) rather than using standardized risk weights.

U.S. broker-dealers, including banks acting as such, are subject to the Securities and Exchange Commission's (SEC's) "net capital rule" (SEC Rule 15c3-1). This rule stipulates, among other things, that a broker must have a minimum "net capital," which is defined as equity capital plus approved subordinate liabilities minus "secu-

rities haircuts” and operational charges. The haircuts are set as security-dependent percentages of the market value. The standard rule requires that the net capital exceeds of least $6\frac{2}{3}\%$ (15:1 leverage) of aggregate indebtedness (broker’s total money liabilities) or alternatively 2% of aggregate debit items arising from customer transactions. This constraint is similar in spirit to (1).⁴ As of August 20, 2004, SEC amended (SEC Release No. 34-49830) the net capital rule for Consolidated Supervised Entities (CSE) such that CSE’s may, under certain circumstances, use their internal risk models similar to (2) to determine whether they fulfill their capital requirement.

2.3 Funding Requirements for Market Makers

There are various types of market-making firms. Some are small partnerships, whereas others are parts of large investment banks. The small firms are financed in a similar way to hedge funds in that they rely primarily on collateralized financing; the funding of banks was described in Section 2.2.

Certain market makers, such as NYSE specialists, have an obligation to make a market and a binding funding constraint means that they cannot fulfill this requirement. Hence, avoiding the funding constraint is especially crucial for such market makers.

Market makers are in principle subject to the SEC’s net capital rule (described in Section 2.2), but the rule has special exceptions for market makers. Hence, market-makers’ main regulatory requirements are those imposed by the exchange on which they operate. These constraints are often similar in spirit to (1).

2.4 Funding Liquidity Risk

Funding liquidity risk is the risk that a trader’s funding constraint is binding. As discussed in Sections 2.1–2.3, the real-world funding constraint is captured by Equation (1), Equation (2), or something in between. In the remainder of this paper, we focus on (1) since it is more realistic in several interesting cases as discussed above.

Funding risk stems from the risks that net capital decreases, short term borrowing availability is reduced, and haircuts increase. Net capital decreases if the institution experiences trading losses or otherwise incurs losses. Short term borrowing can dry up for a bank if it cannot sell commercial paper, or, in the case of a commercial bank, because of demand deposit withdrawals — a bank run. Haircuts increase if the collateral becomes more risky, which happens if fundamental uncertainty increases (e.g. after the crash in 1987 and after September 11, 2001) or if market liquidity is reduced such that liquidation of the collateral is more difficult. For instance, Long Term Capital Management (LTCM) estimated that in times of severe stress, haircuts on AAA-rated

⁴Let L be the lower of $6\frac{2}{3}\%$ of total indebtedness or 2% of debit items, then the rule requires that $L \leq W - \sum_j h^j x^j$, that is, $\sum_j h^j x^j \leq W - L$.

commercial mortgages would increase from 2% to 10%, and similar haircut increases for other securities (HBS Case N9-200-007(A)).

Naturally, financial institutions try to manage their funding liquidity risk. For instance, Goldman Sachs (2003 Annual Report, page 62) states they it seeks to maintain net capital in excess of the total haircuts that it would face in periods of market stress plus the total draws on unfunded commitments at such times. Hence, Goldman Sachs recognizes that it may not have access to short-term borrowing during a crisis, that haircuts may increase during such a crisis, and that counterparties may withdraw funds at such times.

The risk of a funding crisis is not purely academic. For instance, in the 1987 stock market crash numerous market makers hit (or violated) their funding constraint (1).

“By the end of trading on October 19, [1987] thirteen [NYSE specialist] units had no buying power”
— SEC (1988), page 4-58

because of the funding constraint (SEC (1988)). Several of these firms managed to reduce their positions and continue their operations. Others did not. For instance, Tompane was so illiquid that it was taken over by Merrill Lynch Specialists and Beauchamp was taken over by Spear, Leeds & Kellogg (Beauchamp’s clearing broker).

Also, market makers outside the NYSE experienced funding troubles. The Amex market makers Damm Frank and Santangelo were taken over, at least 12 other-the-counter (OTC) market makers ceased operations, and several trading firms ceased operations.

These funding problems were due to (i) reductions in capital arising from trading losses and default on unsecured customer debits; and (ii) an increased funding need stemming from increased inventory and increased haircuts. Haircuts were increased since agents financing the trading perceived that the collateral had an increased risk and a reduced market liquidity. One New York City bank, for instance, increased haircuts from 20% to 25% for certain borrowers, and another bank increased haircuts from 25% to 30% for all specialists (SEC (1988) page 5-27 and 5-28). Other banks reduced the funding period by making intra-day margin calls, and at least two banks made intra-day margin calls based on assumed 15% and 25% losses, thus effectively increasing the haircut by 15% and 25%. Also, some broker-dealers experienced a reduction in their line of credit.

Another stark example of funding liquidity risk is LTCM, which — in spite of their numerous measures to control funding risk — ultimately could not fund its positions and was taken over by 14 banks in September 1998.

3 Model

We consider a discrete-time infinite horizon economy with time indexed by t . There is a riskless security with (gross) interest rate R and J risky securities in zero net supply. Each risky security j pays off its liquidation value v_t^j at maturity $\bar{T}$.⁵ This value follows a random walk with drift such that $v_{t+1}^j = \frac{1}{\delta}v_t^j + \varepsilon_{t+1}^j$ where ε_{t+1}^j are random variables with zero mean, independent across time and support on $[-\Delta_{t+1}, \Delta_{t+1}]$. The bound increases over time, i.e. $\Delta_{t+1} = \frac{1}{\delta}\Delta_t$. All agents learn ε_t^j at the beginning of period t and have no information about future ε 's.

The economy has three kinds of agents (with mass of unity for each type): dealers, initial customers, and late customers as in Grossman and Miller (1988). The initial customers arrive in the market at time t_0 . They choose their holding x_t^j of security j at time $t \geq t_0$ to maximize profit reduced by a quadratic holding cost. That is, the representative initial customer maximizes his utility given by

$$\max_x E \left[\delta^{\bar{T}} w_{T'} - \sum_{t=t_0}^{\bar{T}-1} \delta^t \sum_j \frac{\gamma}{2} (x_t^j + z_t^j)^2 \right] \quad (3)$$

where $\delta = 1/R$ is a discount factor and wealth evolves according to

$$w_{t+1} = R w_t + \sum_j x_t^j (p_{t+1}^j - R p_t^j). \quad (4)$$

The first order condition to the initial customers' problem is

$$0 = \delta E_t(p_{t+1}^j - R p_t^j) - \gamma (x_t^j + z_t^j) \quad (5)$$

so their demand is given by

$$x_t^j = -z_t^j + \kappa E_t(\delta p_{t+1}^j - p_t^j) \quad (6)$$

where $\kappa = 1/\gamma$. These preferences of the initial customers capture the idea that they would like to sell z_t^j shares of asset j if the price is random walk, and they are price sensitive such that their supply curve is upward sloping.

The complementary customers only start trading at time T , where $t_0 < T \leq \bar{T}$, so their demand is $x_t = 0$ for all $t < T''$. The representative complementary customer has utility function

$$\max_x E \left[\delta^{T'} w_{T'} - \sum_{t=T''}^{T'-1} \delta^t \sum_j \frac{\gamma}{2} (x_t^j - z_t^j)^2 \right] \quad (7)$$

We note that the complementary customer's utility function is analog to that of the initial customer, except that the supply shock z_t^j has the opposite side. This means

⁵All random variables are defined on a probability space $(\Omega, \mathcal{F}, \mathcal{P})$.

that the complementary customers are the natural counterparties to the initial customers; they are, however, not in the market at the same time and the dealers provide immediacy by helping to bridge the gap between their arrival times to the market.

The dealers are always in the market and never face (exogenous) holding costs, but, importantly, dealers need to be able to finance their trades. There are μ identical dealers and each dealer solves the following problem:

$$\max_x E \left[\delta^{\bar{T}} w_{\bar{T}} \right]$$

subject to the capital constraint

$$\sum_j (h_t^{j+} x_t^{j+} + h_t^{j-} x_t^{j-}) \leq w_t \quad (8)$$

$$x_t^j = x_t^{j+} - x_t^{j-} \quad (9)$$

$$x_t^{j+} \geq 0 \quad (10)$$

$$x_t^{j-} \geq 0 \quad (11)$$

where $h_t^{j+} = p_t^j - l_t^j$ and $h_t^{j-} = c_t^j - p_t^j$. We assume that the collateralized loans are structured such that they are risk free and, hence, the financing rate is assumed to be the risk-free rate r . More precisely, the loan value of security j at time t is the discounted smallest possible price in the next period. That is, l_t is the largest value such that $\mathcal{P}_t(p_{t+1}^j \geq Rl_t) = 1$ almost surely. Similarly, the shorting collateral is the discounted largest possible price next period. This capital constraint captures the real-world features discussed in Section 2, especially in Section 2.1.

Each dealer is endowed with $x_{t_0-1}^j$ shares of risky security j and b_{t_0} dollars in cash/bonds at the beginning of period t_0 . Hence, their initial wealth is $w_{t_0} = b_{t_0} + \sum_j x_{t_0-1}^j p_{t_0}^j$, which depends on $p_{t_0}^j$ unless $x_{t_0-1} = 0$. The dealer's wealth evolves according to

$$\begin{aligned} w_{t+1} &= R \left[w_t - \left((h_t^+)' x_t^+ + (h_t^-)' x_t^- \right) \right] + \sum_j (x_t p_{t+1}^j - Rl_t x_t^+ + Rc_t x_t^-) \\ &= R w_t + x_t' (p_{t+1} - R p_t) \end{aligned} \quad (12)$$

which is identical to (4) since the dealer's collateralized loans are financed at the risk free rate. We use vector notation for x_t , p_t , h_t , l_t , c_t and note that $W_t = \int w_t(i) di$ and $B_t = \int b_t(i) di$ the aggregate variables of w_t and b_t .

An equilibrium is defined as a price process $\{p_t\}$, which all agents take as given, such that aggregate demand is equal to zero for all risky securities. We next characterize equilibrium and show its liquidity properties.

4 Analysis

At time T when the complementary customers arrive, they naturally take the other side of the initial customers trade since they have the opposite supply shock. Hence, after this time, dealers do not trade and the price is given by the fundamental value as stated in the following lemma.

Lemma 1 *At any time $t \geq T$ when complementary customers have arrived prices equal fundamentals, $p_t^j = v_t^j$.*

We are interested in the price dynamics before time T . We first solve the dealer's problem using dynamic programming for a price process such that $p_t = p_t(A_t)$, where A_t is a state variable. For this, we let $V_t(w_t, A_t)$ be the value function at time t if the dealer has wealth w_t and the state variable is A_t . At time $t = T$, $V_T = w_T = R w_{T-1} + \sum_j x_{T-1}^j (v_T^j - R p_{T-1}^j)$. At any time $t < T$ $w_{t+1} = R w_t + \sum_j x_t^j (p_{t+1}^j - R p_t^j)$ and the value function is

$$V_t(w_t, A_t) = \max_{x_t} \delta E_t [V_{t+1}(w_{t+1}, A_{t+1})], \quad (13)$$

subject to the constraints (8)–(11).

Using the envelope theorem for constrained optimization, we see that

$$\frac{\partial V_t}{\partial w_t} = E_t \left[\frac{\partial V_{t+1}}{\partial w_{t+1}} \right] + \phi_t, \quad (14)$$

where ϕ_t is the Lagrange multiplier for constraint (8) at time t . Iterating this expression with the terminal condition $\frac{\partial V_T}{\partial w_T}$, we get

$$\frac{\partial V_t}{\partial w_t} = 1 + \sum_{\tau=0}^{T-1} E_t [\phi_{t+\tau}] \quad (15)$$

that is, the value of a dollar is its consumption value of 1 at T' plus all current and future expected shadow costs.

To analyze the properties of the model, we first define funding illiquidity and market illiquidity as precise mathematical objects.

Definition 1 (i) *Funding illiquidity is defined as the dealer's marginal value of funds, $\Phi := \frac{\partial V_t}{\partial w_t}$.*

(ii) *Market illiquidity for a trade of size z^j is defined as the absolute value of*

$$\Lambda_t^j(z^j) := v_t^j - p_t^j(z^j). \quad (16)$$

The notion of funding liquidity reflects the marginal value of an extra dollar of funding. The notion of market liquidity is standard. It captures the price impact of a trade or, equivalently the trading cost depending on the trade size.

4.1 Single Period Model

We first discuss how funding liquidity and market liquidity are determined in a single period model. Hence, we assume for now that the initial customers arrive at time $t_0 = 1$ and the complementary customers arrive in the next period $T = 2$. Note that in this case it is never beneficial for $z^j \geq 0$ to short an asset and hence $x_t^j = x_t^{j+} \geq 0$ and $x_t^{j-} = 0$ holds for all assets. The single period setting is also very special in the sense that the loan value $l_1 = v_1 - \Delta$ is independent of future illiquidity ($\Lambda_2 = 0$ by assumption). Hence, the haircut $h_1 = p_1 - l_1$ is simply $\Delta - \Lambda_1$.

The first order condition w.r.t. x_1^j of value function (13) in this case is

$$\Lambda_1^j = \frac{\phi_1}{1 + \phi_1} \Delta_1^j \quad (17)$$

for securites j in which dealers invest ($x_1^j > 0$). That is, the price is

$$p_1^j = \frac{v_1^j + \phi_1 l_1^j}{1 + \phi_1} = v_1^j - \frac{\phi_1}{1 + \phi_1} \Delta_1^j, \quad (18)$$

Equation (18) is intuitive. It shows that prices are closer to fundamentals if funding is less tight (ϕ_1 small), if the collateral value l_1^j is close to the fundamental, or if the fundamental volatility Δ_1^j is small.

In all the remaining assets the dealer do not invest ($x_1^{j'} = 0$). These assets' market illiquidity is given by the initial customers preferences. That is, their marginal holding cost

$$\Lambda_t^{j'} = \gamma z^{j'}.$$

Put differently, the dealer invests all his capital in securities with the maximum expected profit per used capital, $(v_1^j - p_1^j)/(p_1^j - l_1^j)$. The shadow cost of capital, ϕ_1 is the maximum attainable profit per used capital

$$\phi_1 = \max_j \frac{\Lambda_1^j}{\Delta_1^j - \Lambda_1^j} \quad (19)$$

We assume that the dealer's endowment of cash b_1 and shares x_0^j are such that the dealer cannot end up with negative wealth, even if he must choose a zero position, that is,

$$b_1 + x_0' (v_1^j - \gamma z^j) \geq 0 \quad (20)$$

Further, we assume that the initial customers do not want to sell at the price l_1^j , that is, $v_1^j - \gamma z^j > l_1^j$ for all j .

The dealer's capital constraint is

$$b_1 + x_0' p_1^j \geq x_1' (p_1 - l_1) \quad (21)$$

This constraint combined with the equilibrium condition $(x_t = z_t - (v_1 - p_1) / \gamma)$ gives

$$B_1 + x'_0 p_1 \geq (z_t - (v_1 - p_1) / \gamma)' (p_1 - l_1) \quad (22)$$

which determines the equilibrium value of ϕ_1 . Recall that variables x_0, v_1, p_1, l_1 are in vector notation and B is the aggregates of individual b_1 . We prove in appendix that there exists an equilibrium ϕ_1 under our standing assumptions; in case of multiple ϕ_1 that solve (22), we use the lowest one.

We show that the current funding situation for the dealers — as captured by their (possibly negative) cash holding B_1 and the collateral value of securities l_1 — determines the market liquidity of securities as follows:

Proposition 1 (Regions) *There exists a cut-off dealer cash wealth $\underline{B}$ such that*

- (i) *If $B_1 > \underline{B}$ market liquidity is at its maximum and is not sensitive to marginal changes in wealth B_1 and in loan values l_1^k ;*
- (ii) *for $B_1 < \underline{B}$, market liquidity in security j is increasing in wealth B_1 and in loan values l_1^k .*

The results that market liquidity is either “there” or “dried up” is consistent with anecdotal evidence, as well as the finding that the volatility of market liquidity is particularly high during illiquid times. Further, since market making firms are often exposed to market risk, the model implies that market liquidity is low when market makers have low wealth, that is, when the market is down, consistent with empirical evidence. The model further implies that market liquidity is common across securities:

Proposition 2 (Commonality of market liquidity) *If B_1 is random, then for any z^j and z^k , $\text{Cov}(\Lambda_t^j(z^j), \Lambda_t^k(z^k)) > 0$.*

This commonality arises in a straightforward way from the fact that any securities market liquidity depends on the funding condition of the dealer sector.

The model further links market illiquidity to fundamental volatility. To see this, note that if an asset has higher fundamental volatility, then a smaller part of the expected value can be used as collateral because of the inherent risk. Specifically, an asset with higher volatility has a larger value of $v_1^j - l_1^j$. The next proposition demonstrates the volatility implications for market liquidity.

Proposition 3 *Suppose that $B_1 < \underline{B}$ and asset j has higher fundamental volatility than asset k , $\Delta_1^j > \Delta_1^k$. Then*

(i) **(Quality=Liquidity)** *Assets with higher fundamental volatility are more illiquid, that is, $\Lambda^j(z) \geq \Lambda^k(z)$ for any z .*

(ii) **(Flight to Quality)** *If $x_1^j, x_1^k > 0$, then the liquidity differential between high and low fundamental volatility securities is bigger when wealth B_1 is low, that is, $\left| \frac{\partial \Lambda^j}{\partial B_1} \right| > \left| \frac{\partial \Lambda^k}{\partial B_1} \right|$.*

Hence, not only are more volatile securities on average more illiquid (part (i)), they are also more sensitive to changes in dealers' funding condition because dealers' need a high compensation for providing liquidity in high-haircut securities when capital is tight. Empirically, a relation between market liquidity and volatility has been documented by Benston and Hagerman (1974) and Amihud and Mendelson (1989), and flight to liquidity has been documented by Acharya and Pedersen (2005).

When dealers are constrained and enter the time period with a larger position x_0 than they can maintain, a multiplier effect arises. This is because their funding problems lead to attempts to sell, which lowers market liquidity. The lower market liquidity further reduces funding, and so on. Hence, as prices spiral away from fundamentals, the marginal value of funding skyrockets.

This effect is shown in the following proposition in the case of one security $J = 1$. To simplify notation we omit the superscript j . First note that rewriting the constraint (??) as

$$x_1 \leq x_0 + \frac{B_1 + x_0 l_1}{p_1 - l_1}$$

reveals that for $B_1 < -x_0 l_1$ the dealer has to reduce his position (i.e. $x_1 < x_0$), while for $-x_0 l_1 < B_1 < \underline{B}$, the dealer can increase his position ($x_1 > x_0$). Finally, for $B_1 > \underline{B}$ the dealer is unconstrained. Hence, B_1 can fall in one of three regions.

Proposition 4 (Liquidity Spiral) *With $J = 1$, market liquidity is (i) insensitive to marginal shocks if $\underline{B} < B_1$, (ii) moderately sensitive if $-x_0 l_1 < B_1 < \underline{B}$, or (iii) highly sensitive through a multiplier effect that depends on the dealers' leverage $p_1 - l_1$ if $B_1 < -x_0 l_1$. More precisely, the following comparative statics apply in the three respective regions for cash wealth B_1 :*

B_1	$\partial x_1 / \partial B_1$	$\partial p_1 / \partial B_1$	$\partial x_1 / \partial v_1$	$\partial p_1 / \partial v_1$	$\partial x_1 / \partial z_1$	$\partial p_1 / \partial \gamma z_1$
$< -x_0 l_1$	$> 1/h_1$	$> \gamma/h_1$	$> x_0/h_1$	$> 1 + \frac{\gamma x_0}{h_1}$	< 0	< -1
$\in (-x_0 l_1, \underline{B})$	$< 1/h_1$	$< \gamma/h_1$	$\in (0, x_0/h_1)$	$\in (1, 1 + \frac{\gamma x_0}{h_1})$	$\in (0, 1)$	$\in (-1, 0)$
$> \underline{B}$	0	0	0	1	1	0

The last line of the Proposition 4 shows that if funding is abundant, i.e. $B_1 > \underline{B}$, a marginal change in B_1 has no implications and a larger z_1 shock is fully absorbed by the dealers. For the constrained cases with $B_1 < \underline{B}$ let us first focus on the comparative static with respect to a larger z_1 shock (last two columns). Recall the initial customer's supply curve is

$$s_1 = z_1 - \{v_1 - p_1\} / \gamma = -\{v_1 - p_1 - y_1\} / \gamma, \quad (23)$$

where $y^j = \gamma z_1^j$ is the supply shock in dollars rather than in shares. Graphically, increasing z_1 by Δz_1 shifts the supply curve parallel to the right by Δz_1 (or vertically downwards by $\gamma \Delta z_1$). The multiplier effect arises for $B_1 < -x_0 l_1$ when dealers have to reduce their position, i.e. $x_1 < x_0$. An increase in the supply shock z leads to a

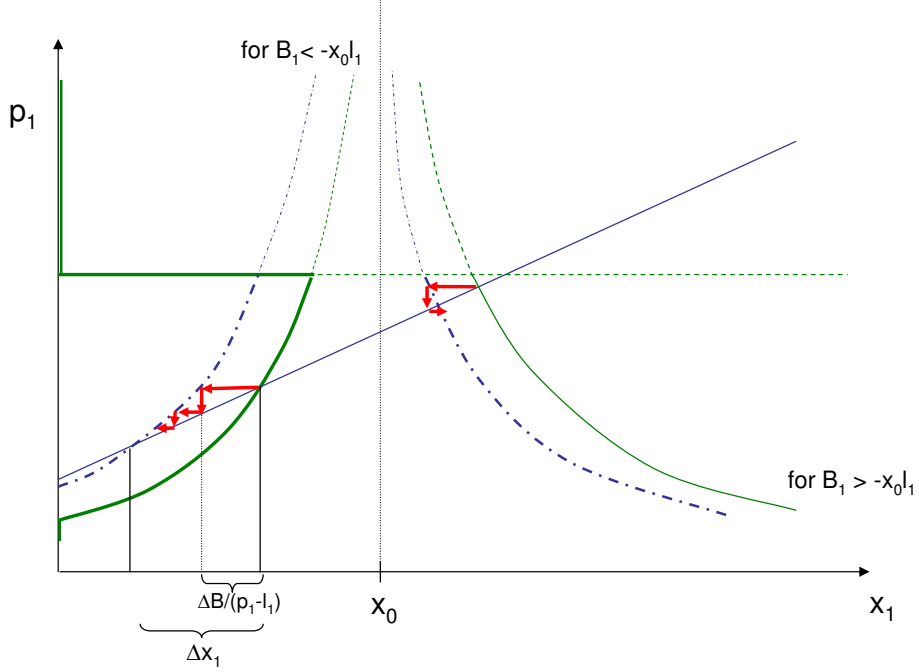


Figure 1: Dealers' demand is backward bending if $B_1 < -x_0 l_1$. This leads to an amplification of price shocks.

reduction in price, which leads to dealer losses on the existing positions x_0 , leading to a further price drop, etc. forcing the constrained dealer to reduce his position x_1 even further. Hence, an increased demand for immediacy leads to a reduction in the provision of immediacy. Said differently, the dealer's demand curve is backward bending when he is constrained and has a large position that he cannot roll over. This also explains, why $\partial p / \partial(\gamma z_1)$ the price drops by more than γz_1 . For $B_1 \in (-x_0 l_1, \underline{B})$, the larger z_1 shock is partially absorbed by the dealers and hence the price shock is smaller than $\gamma \Delta z_1$.

Next, we analyze the comparative statics with respect to a slightly smaller (or more negative) bond holding. Reducing B_1 forces dealers with $B_1 < -x_0 l_1$ to sell more shares. For each share sold, they raise only $p_1 - l_1$ extra dollars since their positions are leveraged. Moreover, their selling depresses the price which erodes their current wealth $B_1 + x_0 p_1^j$ even further, forcing them to sell even more and depressing the price even further. This leads to a multiplier effect illustrated in Figure 4.1. Formally, $\partial x_1^j / \partial B_1 > 1 / (p_1 - l_1)$. If $B_1 \in (-x_0 l_1, \underline{B})$, dealers acquire shares, i.e. $x_1 > x_0$. In this case, lowering B_1 reduces their funds to purchase shares and to provide liquidity. This lowers the equilibrium price which partially mitigates the demand effect, rather than creating a multiplier as in the previous case.

Hence, in both cases market liquidity and funding liquidity are mutually reinforcing, but when $B_1 < -x_0 l_1$ market liquidity is erratic due to the multiplier effect. Interestingly, the larger is the dealer's leverage, the larger is the multiplier effect.

Finally, let us analyze a drop in the (discounted) fundamental value of the stock v_1 by Δ . This comparative statics exercise is of particular interest for our subsequent dynamic setting. The decline in fundamental value shifts the supply curve of the initial customers down by Δ . Similarly, the horizontal line of dealers' demand curve moves down by Δ . Hence, for $B_1 > \underline{B}$ the price simply drops by Δ as well and x_1^j remains unchanged. Note that l_t also drops by Δ . Now, suppose that p_1^j also drops by Δ in the constrained cases, then the funding constraint $x_1 \leq x_0 + (B_1 + x_0 l_1) / (p_1 - l_1)$ becomes more binding. Put differently, in order to obtain the same x_1 , one had to increase the dealers bond holding B_1 by $x_0 \Delta$. Hence, a decline in v_1 by Δ reduces x_1 and causes a price drop larger than Δ for $B_1 < \underline{B}$. As discuss above, if $B_1 < -x_0 l_1$ then a multiplier effect arises, which makes market liquidity erratic.

The insights of the single asset case can be extended to a setting with multiple assets. (For the time being) we provide only a brief outline for this generalization. Recall that the risk-neutrality of the dealers implies that dealers only invest in the assets with the highest attainable profit per used capital $\phi_1 = \max_j (v_1 - p_1^j) / (p_1^j - l_1^j)$ and nothing in the other assets. Since in any equilibrium dealers have to be indifferent among the assets within the first group, their relative prices are pinned down. Hence, we can treat this group of assets as a single bundle and apply for this bundle the same reasoning as for the single asset in the single asset case.⁶

4.2 Dynamic Model: Liquidity Risk and Central Bank Policy

In this section we consider a dynamic model of liquidity provision. An interesting feature of the dynamic set-up is that dealers' hair cuts depend on future prices, which depend on future market liquidity, which in turn depend on future hair cuts, and so on. Hence, there is a dynamic connection between market and funding liquidity.

More formally, the haircut h_t at time t depends on current and future illiquidity as follows

$$h_t = p_t - l_t \tag{24}$$

$$= v_t - \Lambda_t - \delta \min\{v_{t+1} - \Lambda_{t+1}\} \tag{25}$$

$$= \Delta_t + \max \delta \Lambda_{t+1} - \Lambda_t \tag{26}$$

The haircut is the sum of fundamental risk as captured by Δ and liquidity risk as captured by $\max \delta \Lambda_{t+1} - \Lambda_t$. Hence, if future liquidity worsens — that is, if $\max \delta \Lambda_{t+1}$ is higher — then, ceteris paribus, current haircuts increase and funding worsens.

⁶In the formal proof we also have to consider that the composition of the bundle might change.

The market liquidity Λ_t^j of asset j depends on other assets' market liquidity and overall funding liquidity. First, we show that the risk that future funding constraints might bind – funding liquidity risk – can reduce current market liquidity provision even when the current funding constraint is not binding. Second, we focus on the cross-sectional liquidity provision across different assets. The final part of this subsection outlines some Fed Policies.

Liquidity-Reducing Liquidity Risk. The dealers' first order condition (Euler equation)

$$p_t^j = E_t \left[\delta \frac{\Phi_{t+1}}{\Phi_t} p_{t+1}^j \right] \quad (27)$$

shows that the stochastic discount factor, $\delta \frac{\Phi_{t+1}}{\Phi_t}$, is next period's funding illiquidity divided by current funding illiquidity times δ .

For assets with $p_t^j \leq v_t^j$ for all t (i.e. for assets with positive and sufficiently large z -shock), $\Lambda_t^j = v_t^j - p_t^j$. Hence, equation (27) in terms of market illiquidity is

$$\Lambda_t^j = \delta E_t [\Lambda_{t+1}^j] E_t \left[\frac{\Phi_{t+1}}{\Phi_t} \right] - \delta Cov_t \left[\frac{\Phi_{t+1}}{\Phi_t}, p_{t+1}^j \right]. \quad (28)$$

For assets with $p_t^j \geq v_t^j$ for all t , $\Lambda_t^j = p_t^j - v_t^j$, and we have to modify the equation by adding the covariance terms instead of subtracting it.⁷

Note that in the dynamic case, abundance of funding at t does not guarantee perfect liquidity provision at time t . Dealers may restrict their current market liquidity provision because funds might be scarce in some future state. If the funding constraint does not bind at t , $\phi_t = 0$ and

$$\Lambda_t^j = \delta E_t [\Lambda_{t+1}^j] - \frac{\delta Cov_t [\Phi_{t+1}, p_{t+1}^j]}{E_t [\Phi_t]}. \quad (29)$$

This equation reveals that potentially binding future funding constraints alone can lead to current market illiquidity for two reasons. First, Λ_t^j is non-zero since funding constraints at $t + 1$, possibly amplified by a multiplier effect, lead to positive expected Λ_{t+1}^j . Second, the market liquidity may be even further depressed if p_{t+1}^j is negatively correlated with next period's funding illiquidity Φ_{t+1} . The second effect arises since dealers manage a “buffer.” They hold back some capital for future states of the world in which extra funding is very valuable. In these states the market is illiquid and it is very costly to shed shares, and very cheap to acquire additional shares. Reducing their

⁷Note that in the special case with constant funding liquidity Φ_t over time, $\Lambda_t^j = \delta E_t [\Lambda_{t+1}^j]$. Constant funding liquidity can for example be achieved in a setting in which complementary customers arrive stochastically with a constant arrival probability α and dealers are perfectly hedged (i.e. their wealth does not vary over time).

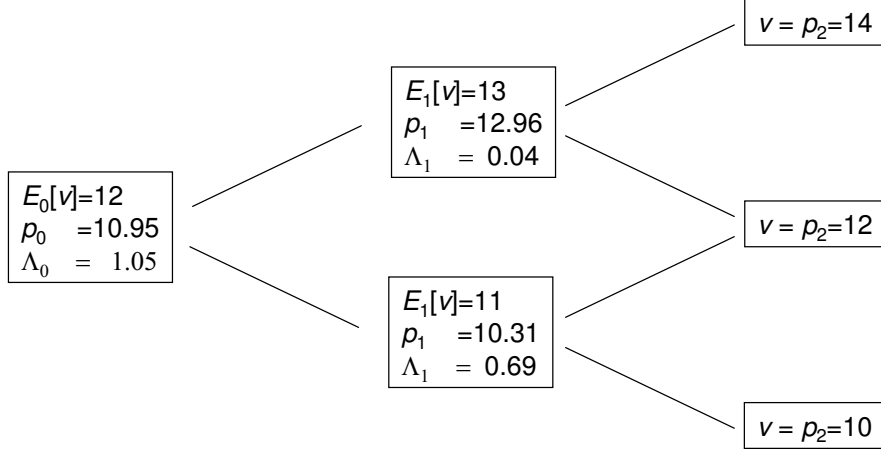


Figure 2: Event tree of fundamentals, prices, and market liquidity for parameter values $\delta = 1/R = 1$, $\mu = 1$, $\kappa = 1$, $z = 1$, and $W_1 = .3$.

market liquidity provision in $t = 0$ lowers the number of shares dealers are forced to sell, or allows them to cheaply purchase shares and to benefit from a liquidity crisis. For this reason they may not make full use of their available funds at time t , even though dealers are risk-neutral and the expected price increase exceeds the risk-free interest rate.⁸ Shleifer and Vishny (1997) provide a similar result within a setting with performance-based arbitrage funding.

Figure 2 numerically quantifies the effects in a two-period example with $t_0 = 0$ and $T = 2$. Fundamentals of a single asset follow a symmetric random walk. The top number at each node states the fundamental value, which would be the prevailing price if there were no funding constraints. We set $\delta = 1/R = 1$, $\mu = 1$, $\kappa = 1$, $z = 1$, and $W_1 = .3$.

Our numerical results show that market is even less liquid at $t = 0$, $\Lambda_0 = 1.05$, than in the down state in $t = 1$, $\Lambda_1^{\text{down}} = 0.69$. Market illiquidity in the up state is $\Lambda_1^{\text{up}} = 0.04$. Further we note that even though the expected price in $t = 1$, $E_0[p_1] = (12.96 + 10.31)/2 = 11.63$ exceeds $p_0 = 10.95$, dealers only use roughly two thirds of their available capital in $t = 0$. They only buy 0.32 shares in $t = 0$ because they are aware that they are forced to reduce their position to 0.31 shares in the bad state in $t = 1$. They are reluctant to be more aggressive in $t = 0$, since $p_1^{\text{down}} = 10.31$ is close to $l_1^{\text{down}} = 10$. This makes it very costly to sell shares in $t = 1$ (in particular since selling raises only additional funds of $p_1^{\text{down}} - l_1^{\text{down}}$ dollars per share) or equivalently

⁸As we will see below, the second effect goes in the opposite direction for assets whose price increases in the “stress scenario state” (i.e. $\text{Cov}_0[\Phi_1, p_1^j] > 0$).

very cheap to acquire additional shares then because the leverage ratio is high. In this state the value of an extra dollar of funding is $\phi_1^{\text{down}} = \frac{11-10.31}{10.31-10} = 2.23$.

Liquidity Provision in the Cross Section. To derive the cross-sectional implication we split the covariance term of equation (28) into two terms: a fundamental term and a liquidity term.

$$\Lambda_t^j = \delta E_t [\Lambda_{t+1}^j] E_t \left[\frac{\Phi_{t+1}}{\Phi_t} \right] - Cov_t \left[\frac{\Phi_{t+1}}{\Phi_t}, \delta v_{t+1}^j \right] + Cov_t \left[\frac{\Phi_{t+1}}{\Phi_t}, \delta \Lambda_{t+1}^j \right]$$

The first covariance term shows that assets whose fundamental increases when funding becomes tight are more liquid. One can easily show that for assets with $p_t^j \geq v_t^j$ (i.e. with significantly negative z^j -shock) one has to add the first covariance term instead of subtracting it. The second covariance term reveals that assets whose future market liquidity is positively correlated with funding liquidity are less liquid. This effect is independent of whether the initial z^j -shock was positive or negative.

We can demonstrate the main effects in our two period setting with $t_0 = 0$ and $T = 2$. For this case we know that $\Lambda_1 = \frac{\phi_1}{1+\phi_1} \Delta_1^j$ for all assets in which dealers invests. Hence, if the funding constraint is not binding in $t = 0$,

$$\begin{aligned} \Lambda_0^j &= \delta E_0 [\Lambda_1^j] - Cov_0 \left[\frac{\Phi_1}{E_0 [\Phi_1]}, \delta v_1^j \right] + \Delta_1^j \delta \frac{1}{E_0 [\Phi_1]} Cov_0 \left[\Phi_1, \frac{-1}{\Phi_1} \right], \\ &= \delta E_0 [\Lambda_1^j] - Cov_0 \left[\frac{\Phi_1}{E_0 [\Phi_1]}, \delta v_1^j \right] + \Delta_1^j \delta \left\{ E_0 \left[\frac{1}{\Phi_1} \right] - \frac{1}{E_0 [\Phi_1]} \right\} \end{aligned}$$

Note that the last term increases in the fundamental volatility Δ_1^j . This replicates our “quality=liquidity result” of Proposition 3 for the dynamic setting. By Jensen’s inequality the term in the curly brackets is always positive and increasing with the variability of funding liquidity Φ_1 . That is, if the shadow costs of an extra dollar of funding fluctuates a lot – like in times of crisis – then markets are relatively less liquid.

The first covariance term states that assets whose fundamental increases when funding is scarce are more liquid. The opposite is true for assets with $p_t^j \geq v_t^j$ (i.e. with large negative z^j -shocks) because the dealer sector is short these assets. Shorting an asset whose correlation between the fundamentals and funding liquidity is positive withdraws money exactly when it is needed the most.

Finally, we show that the dealer sector might even withdraw liquidity from assets. For this purpose we return to our numerical example from above. Let us, however, introduce a second asset. For simplicity let us assume that the fundamentals of both assets are identical and hence perfectly correlated. The assets only differ in the z -shocks that initial customers face. Let the z -shock of asset 1 be one and of asset 2 be tiny, say ε . In $t = 1$ no dealer commits any of his scarce capital to the second asset. Hence,

the prices of asset 2 will almost coincide with the fundamentals, i.e. $p_1^{2,\text{up}} = 13 - \gamma\varepsilon$ and $p_1^{2,\text{down}} = 11 - \gamma\varepsilon$. Are dealers also inactive at time $t = 0$ and hence, is the second asset's price simply $12 - \gamma\varepsilon$? Recall that dealers are especially funding constrained in the down state at time $t = 1$, since at that point investing in the first asset is a great profit opportunity. The obvious way to transfer funds to this state is to go short asset 2 in $t = 0$. This makes it optimal for the dealers to short asset 2 in $t = 0$, which relaxes the funding constraint in the down state in $t = 1$ but also pushes its price below $12 - \gamma\varepsilon$. Said differently, dealers “withdraw” liquidity of asset 2 in $t = 0$.

Central Bank Policy. A central bank can affect market liquidity indirectly by boosting the funding of dealers. In the US, the Fed only lends to commercial banks, but sometimes the Fed “asks” commercial banks to extend credit to securities firms. Hence, the Fed can improve the dealers’ funding by ensuring that they get loans (i.e. increase W) or by ensuring that they get reduced haircuts.

The Federal Reserve Bank of New York (FRBNY) did just that during the 1987 crash:

“calls were placed by high ranking officials of the FRBNY to senior management of the major NYC banks, indicating that ... they should encourage their Wall Street lending groups to use additional liquidity being supplied by the FRBNY to support the securities community”
— SEC (1988), page 5-25

Importantly, the central bank can improve liquidity by stating an intention to fund dealers if needed in the future, without actually taking immediate action. To see this, note that haircuts depend on $\max \Lambda_{t+1}$, and — if the central bank can credibly signal that it will improve funding in a crisis — then future illiquidity expectation $\max \Lambda_{t+1}$ decreases. This, in turn, reduces current haircuts, thus increasing current funding and market liquidity.

The Federal Reserve also used this implicit channel to further improve liquidity in 1987.

“the statement issued by the Chairman of FRB indicating that the FRB would be ready ‘to serve as a source of liquidity to support the economy and financial system’ was considered significant.”
— SEC (1988), page 5-27

5 Related Literature

In this section, we review how our paper links the literatures on market liquidity and funding liquidity.

The market liquidity literature shows that a security can be costly to trade — that is, has less than perfect market liquidity — because of, exogenous order processing costs, private information (Kyle (1985) and Glosten and Milgrom (1985)), inventory risk of market makers (e.g. Stoll (1978), Ho and Stoll (1981,1983) Ho and Stoll (1981) and Grossman and Miller (1988)), search frictions (Duffie, Gârleanu, and Pedersen (2003, 2003a)), or predatory trading (Brunnermeier and Pedersen (2005)). This literature assumes that intermediaries have no funding constraints, with a few exceptions. Attari, Mello, and Ruckes (2005) and Brunnermeier and Pedersen (2005) consider the strategic interactions of large traders who may face funding problems and market-liquidity reducing predation. In Vayanos (2004) liquidity premia are time varying because costly withdrawal of funds are more likely during volatile times. Weill (2004) assumes a position limit for market makers. Our model of market liquidity is similar to Grossman and Miller (1988), but we complement this literature by endogenizing the funding of intermediaries, and linking this funding problem to a unified explanation for the time variation and cross-sectional variation of market liquidity.

Empirically, Chordia, Roll, and Subrahmanyam (2000), Hasbrouck and Seppi (2001) and Huberman and Halka (2001) document that there is commonality of stocks’ market liquidity, that is, market liquidity is correlated across stocks.⁹ Our model shows that this commonality in market liquidity can be driven by the underlying funding liquidity of the market making sector. Acharya and Pedersen (2005) find that when aggregate market liquidity falls, it falls primarily for illiquid assets — a notion often termed “flight to liquidity.” In our model, flight to liquidity arises when dealers have funding problems. This is because dealers prefer to use their limited capital in liquid markets with small haircuts.

Chordia, Sarkar, and Subrahmanyam (2005) show that increases in volatility lead to reductions in market liquidity of bonds and stocks. This is consistent with our model since haircuts should rise with volatility. Chordia, Sarkar, and Subrahmanyam (2005) further document that, “during crisis periods, monetary expansions are associated with increased [market] liquidity.” Our model predicts that monetary policy does not affect liquidity during times of abundant funding of market makers, but that easing the funding restrictions of market makers through monetary policy improves liquidity during crisis.

The second liquidity concept, funding liquidity, is studied in the corporate finance literature. This literature explains how funding liquidity problems arise because of agency problems combined with contract and market incompleteness (see e.g. Hart (1995) and references therein). Similarly, in our model, agents can only collateralize part of the value of an asset, that is, there are positive haircuts.

⁹Market liquidity also affects the overall price level. Amihud and Mendelson (1986) show that assets with a low level of liquidity have on average a higher return. Pastor and Stambaugh (2003) and Acharya and Pedersen (2005) document that aggregate time variation in market liquidity appears to be priced.

The banking literature emphasizes funding liquidity shocks of banks due to early withdrawal by their customers. The role of banks is to insure across customers' time preference shocks by investing in projects and assets with the right maturity structure. However, as Bryant (1980) and Diamond and Dybvig (1983) show, banks are subject to bank-runs if they offer demand deposit contracts (and markets are incomplete). In the case of a bank-run, they are forced to liquidate long-run investment projects at a low liquidation value. Most of the banking literature follows Diamond and Dybvig (1983) in assuming an exogenous liquidation technology — that is, market liquidity is not endogenized. In recent work, however, Allen and Gale (2004) consider a banking model in which the sell-off of the bank's long-run asset may depress its liquidation value. Similarly, Shleifer and Vishny (1992) show within a corporate finance context that a funding crisis can lead to “fire sales” of machines since potential buyers of these industry-specific machines also face similar funding problems at the same time. Our model applies the corporate finance insights of Shleifer and Vishny (1992) and Allen and Gale (2004) to study endogenous market liquidity when the market making sector faces hair-cut constraints. Our multi-asset model also allows us to derive cross-sectional asset price implications. Holmström and Tirole's (1998, 2001) research focuses primarily on funding liquidity. They show that corporations with agency problems have a preference for government bonds because they provide a cushion for future funding liquidity problems. Hence, government bonds trade at a premium.

Our paper is also related to parts of the literature on the “limits to arbitrage.” Shleifer and Vishny (1997) show, among other things, that a demand shock can be amplified if losses lead to withdrawal of capital from fund managers. We show that a similar effect can arise due to leverage and document how the multiplier is exacerbated by the degree of leverage (Proposition 4). Liu and Longstaff (2004) derive the optimal dynamic arbitrage strategy under funding constraints in a setting with exogenous price process. Gromb and Vayanos (2002) derive welfare results in a model in which arbitrageurs face the haircut constraints used in our paper. They also show that arbitrageurs are unconstrained if the initial wealth is large enough, and constrained otherwise. Their focus on welfare and the case of no aggregate risk is different from ours. Our wealth effects can also be related to certain work on contagion (see e.g. Brunnermeier and Pedersen (2005), Kyle and Xiong (2001)).¹⁰

Finally, our paper also touches on the literature that studies the amplification mechanism over the business cycle due to agency costs. This literature was initiated by Bernanke and Gertler (1989) who show that higher net wealth during booms reduces agency costs, while the lower net wealth during recessions leads to higher agency costs and tighter collateral constraints (see also Eisfeldt (2004)). Kiyotaki and Moore (1997) show in a deterministic model how a temporary productivity shock can lead to a dynamic multiplier. This happens because the shock lowers the net worth of “farmers,”

¹⁰See also DeLong, Shleifer, Summers, and Waldmann (1990) and Abreu and Brunnermeier (2002).

thus constraining their production, which reduces their future net worth, and so on. This reduction in the farmers' current and future demand for the asset reduces its price. We consider a model with uncertainty and multiple assets, which enables us to study the interaction between risk and credit constraints, and cross-sectional implications such as flight to quality.

A Proofs

Proof of Proposition 1. First, if the funding constraint is not binding then market liquidity is at its maximum in the sense that the expected return of every security is equal to the risk free return. In this case, prices are insensitive to marginal changes in funding conditions, i.e. to B_1 and l_1^k .

If the constraint is binding then clearly market liquidity is lower. Market liquidity is tied to ϕ_1 through equation (??). Further, $\frac{\partial \phi_1}{\partial B_1} < 0$. To see this, consider, respectively, the right-hand side (rhs) and left-hand side (lhs) of (??) as functions of ϕ_1 , where p_1 depends on ϕ_1 as in (??). Since the constraint is binding, $\text{lhs} < \text{rhs}$ at $\phi_1 = 0$. Further, both the lhs and rhs are decreasing continuous functions of ϕ_1 . As ϕ_1 increases, p_1^j decreases until it becomes equal to its lower bound $\delta E_1(v^j) - \frac{z^j}{\kappa}$. When ϕ_1 increases to some value ϕ such that all prices have reached their lower bounds then the rhs is zero because $x_1^j = 0$ for all j . Further, the lhs is non-negative because of our assumption of the solvency of the dealer (??). Hence, there exists an equilibrium ϕ_1 smaller than or equal to ϕ . If B_1 is smaller then the lhs is smaller and the (first) intersection of the lhs and rhs happens for a larger ϕ_1 . Similarly, if l_1^k is larger, the rhs is lower and the (first) intersection happens for a lower value of ϕ_1 . ■

Proof of Proposition 2. Recall that $\frac{\partial p_1^j}{\partial B_1} > 0$ for any asset j for $B_1 < \underline{B}$. The results follows from the fact that two functions which are both increasing in same random variable are positively correlated. ■

Proof of Proposition 3. (i) It can be seen that if $x_1^k = 0$ then $x_1^j = 0$ and hence each security is equally illiquid. If $x_1^j = 0$ and $x_1^k > 0$ then clearly $\Lambda^j(z) \geq \Lambda^k(z)$. Finally, if $x_1^j > 0$, then $x_1^k > 0$, and for $m = j, k$,

$$\Lambda^m(z) = -(p_1^m - \delta E_1[v^m]) = \frac{\phi_1}{1 + \phi_1} H_1^m$$

Hence, $\Lambda^j(z) \geq \Lambda^k(z)$.

(ii) For $m = j, k$ we have

$$\left| \frac{\partial \Lambda^m(z)}{\partial B_1} \right| = \left(\frac{\partial}{\partial \phi_1} \frac{\phi_1}{1 + \phi_1} \right) \left| \frac{\partial \phi_1}{\partial B_1} \right| H_1^m$$

which clearly increases in H_1^m . ■

Proof of Proposition 4. If $B_1 > \underline{B}$, then $p_1^j = E_1[v^j]$ from which the results follow.

For $B_1 \leq \underline{B}$,

$$B_1 + p_1 x_0 = (p_1 - l_1) x_1$$

which can be rewritten as

$$x_1 = x_0 + \frac{B_1 - x_0 l_1}{p_1 - l_1}.$$

Hence, $x_1 < x_0$ iff $B_1 < -l_1 x_0$.

Part 1: Derivatives w.r.t. $y := z_1/\kappa$.

The constraint $B_1 + p_1 x_0 = (p_1 - l_1)x_1$ implies that

$$\partial x_1 / \partial y = \frac{x_0 - x_1}{p_1 - l_1} \partial p_1 / \partial y.$$

Recall the equilibrium condition $z_t^j - \kappa \{ \delta E_t [p_{t+1}] - p_t^j \} = \mu_t x_t^j$, which implies that $\partial p_1 / \partial z = -\frac{1}{\kappa} + \frac{\mu}{\kappa} \frac{\partial x}{\partial z}$ or, equivalently,

$$\frac{\partial p_1}{\partial y} = -1 + \frac{\mu}{\kappa} \frac{\partial x_1}{\partial y}$$

Combining these equations yields

$$\frac{\partial p_1}{\partial y} = -1 + \frac{\mu}{\kappa} \frac{x_0 - x_1}{p_1 - l_1} \frac{\partial p_1}{\partial y}$$

that is,

$$\frac{\partial p_1}{\partial y} = \frac{-1}{1 - \frac{\mu}{\kappa} \frac{x_0 - x_1}{p_1 - l_1}}$$

which is small than minus one if $x_1 < x_0$. (If an equilibrium exists then the denominator is positive.)

Part 2: Derivatives w.r.t. B_1 .

The equilibrium condition implies that $p_1 = \delta E_1 [v] + \frac{1}{\kappa} (\mu x_1 - z_1)$. This, together with the funding constraint, leads to

$$\frac{\partial x_1}{\partial B_1} = \frac{1}{(p_1 - l_1) + \frac{\mu}{\kappa} (x_1 - x_0)}.$$

which is greater than $\frac{1}{p_1 - l_1}$ if $x_1 < x_0$ and smaller otherwise.

From the equilibrium condition follows immediately that

$$\frac{\partial p_1}{\partial B_1} = \frac{\mu}{\kappa} \frac{\partial x_1}{\partial B_1} = \frac{1}{\frac{\kappa}{\mu} (p_1 - l_1) + (x_1 - x_0)}.$$

Part 3: Derivatives w.r.t. fundamental value $E_1 [v]$.

The equilibrium and funding conditions imply that

$$\frac{\partial x_1}{\partial \delta E_1(v)} = \frac{x_0}{p_1 - l_1 + \frac{\mu}{\kappa} (x_1 - x_0)}$$

and that

$$\frac{\partial p_1}{\partial \delta E_1(v)} = 1 + \frac{\mu}{\kappa} \frac{\partial x_1}{\partial \delta E_1(v)}.$$

■

References

- ABREU, D., AND M. K. BRUNNERMEIER (2002): “Synchronization Risk and Delayed Arbitrage,” *Journal of Financial Economics*, 66(2-3), 341–360.
- ACHARYA, V. V., AND L. H. PEDERSEN (2005): “Asset Pricing with Liquidity Risk,” *Journal of Financial Economics* (forthcoming).
- ALLEN, F., AND D. GALE (2004): “Financial Intermediaries and Markets,” *Econometrica*, 72(4), 1023–1061.
- AMIHUD, Y., AND H. MENDELSON (1986): “Asset Pricing and the Bid-Ask Spread,” *Journal of Financial Economics*, 17(2), 223–249.
- (1989): “The Effects of Beta, Bid-Ask Spread, Residual Risk, and Size on Stock Returns,” *Journal of Finance*, 44(2), 479–486.
- ATTARI, M., A. S. MELLO, AND M. E. RUCKES (2005): “Arbitraging Arbitrageurs,” *Journal of Finance* (forthcoming).
- BENSTON, G. J., AND R. L. HAGERMAN (1974): “Determinants of the Bid-Ask Spread in the Over-the-Counter Market,” *Journal of Financial Economics*, 1(4), 353–364.
- BERNANKE, B., AND M. GERTLER (1989): “Agency Costs, Net Worth, and Business Fluctuations,” *American Economic Review*, 79(1), 14–31.
- BRUNNERMEIER, M. K., AND L. H. PEDERSEN (2005): “Predatory Trading,” *Journal of Finance* (forthcoming).
- BRYANT, J. (1980): “A Model of Reserves, Bank Runs, and Deposit Insurance,” *Journal of Banking and Finance*, 4, 335–344.
- CHORDIA, T., R. ROLL, AND A. SUBRAHMANYAM (2000): “Commonality in Liquidity,” *Journal of Financial Economics*, 56, 3–28.
- CHORDIA, T., A. SARKAR, AND A. SUBRAHMANYAM (2005): “An Empirical Analysis of Stock and Bond Market Liquidity,” *Review of Financial Studies*, 18(1), 85–129.
- DELONG, J. B., A. SHLEIFER, L. H. SUMMERS, AND R. J. WALDMANN (1990): “Noise Trader Risk in Financial Markets,” *Journal of Political Economy*, 98(4), 703–738.
- DIAMOND, D., AND P. DYBVIG (1983): “Bank Runs, Deposit Insurance, and Liquidity,” *Journal of Political Economy*, 91(3), 401–419.

- DUFFIE, D., N. GÂRLEANU, AND L. H. PEDERSEN (2002): “Securities Lending, Shorting, and Pricing,” *Journal of Financial Economics*, 66(2-3), 307–339.
- (2003): “Valuation in Over-the-Counter Markets,” *Working Paper, Stanford University*.
- EISFELDT, A. (2004): “Endogenous Liquidity in Asset Markets,” *Journal of Finance*, 59(1), 1–30.
- GLOSTEN, L. R., AND P. R. MILGROM (1985): “Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders,” *Journal of Financial Economics*, 14, 71–100.
- GROMB, D., AND D. VAYANOS (2002): “Equilibrium and Welfare in Markets with Financially Constraint Arbitrageurs,” *Journal of Financial Economics*, 66(2-3), 361–407.
- GROSSMAN, S. J., AND M. H. MILLER (1988): “Liquidity and Market Structure,” *Journal of Finance*, 43(3), 617–633.
- HART, O. D. (1995): *Firms, Contracts, and Financial Structure*. Oxford University Press, Oxford, (Clarendon Lectures in Economics).
- HASBROUCK, J., AND D. SEPPI (2001): “Common Factors in Prices, Order Flows and Liquidity,” *Journal of Financial Economics*, 59(2), 383–411.
- HO, T. S. Y., AND H. R. STOLL (1981): “Optimal Dealer Pricing Under Transactions and Return Uncertainty,” *Journal of Financial Economics*, 9, 47–73.
- (1983): “The Dynamics of Dealer Markets under Competition,” *Journal of Finance*, 38(4), 1053–1074.
- HOLMSTRÖM, B., AND J. TIROLE (1998): “Private and Public Supply of Liquidity,” *Journal of Political Economy*, 106(1), 1–39.
- (2001): “LAPM: A Liquidity-Based Asset Pricing Model,” *Journal of Finance*, 56(5), 1837–1867.
- HUBERMAN, G., AND D. HALKA (2001): “Systematic Liquidity,” *Journal of Financial Research*, 24, 161–178.
- KIYOTAKI, N., AND J. MOORE (1997): “Credit Cycles,” *Journal of Political Economy*, 105(2), 211–248.
- KYLE, A. S. (1985): “Continuous Auctions and Insider Trading,” *Econometrica*, 53, 1315–1335.

- KYLE, A. S., AND W. XIONG (2001): “Contagion as a Wealth Effect,” *Journal of Finance*, 56(4), 1401–1440.
- LIU, J., AND F. A. LONGSTAFF (2004): “Losing Money on Arbitrages: Optimal Dynamic Portfolio Choice in Markets with Arbitrage Opportunities,” *Review of Financial Studies*, 17(3), 611–641.
- PASTOR, L., AND R. F. STAMBAUGH (2003): “Liquidity Risk and Expected Stock Returns,” *Journal of Political Economy*, 111(3), 642–685.
- SHLEIFER, A., AND R. W. VISHNY (1992): “Liquidation Values and Debt Capacity: A Market Equilibrium Approach,” *Journal of Finance*, 47(4), 1343–1366.
- (1997): “The Limits of Arbitrage,” *Journal of Finance*, 52(1), 35–55.
- STOLL, H. R. (1978): “The Supply of Dealer Services in Securities Markets,” *Journal of Finance*, 33(4), 1133–1151.
- VAYANOS, D. (2004): “Flight to Quality, Flight to Liquidity and the Pricing of Risk,” Working paper, LSE.
- WEILL, P.-O. (2004): “Leaning Against the Wind,” *Mimeo, New York University*.